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Date: 12 December 2025

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Effect of welding speed on keyhole dynamic behavior and penetration depth evolution during pulsed laser welding of titanium alloy

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Abstract

Due to the advantage of precise heat input control, pulsed laser welding has been widely applied in the manufacture of titanium alloy missile grid fins. Compared with continuous laser welding, pulsed laser welding presented a more complex flow behavior of molten pool, which brought significant challenges to the online control of penetration depth. This study established a three-dimensional numerical simulation model of the molten pool flow field and temperature field based on actual measured keyhole geometry size, aiming to reveal the evolution law of the keyhole morphology and its coupling relationship with penetration depth during pulsed laser welding of titanium alloys. It was indicated that the morphology of molten pool was closely related to the penetration depth. With the increase of welding speed, the heat diffusion pattern caused by laser irradiation

Outline

Abstract

Graphical abstract

Keywords

1. Introduction

2. Experimental setup and method

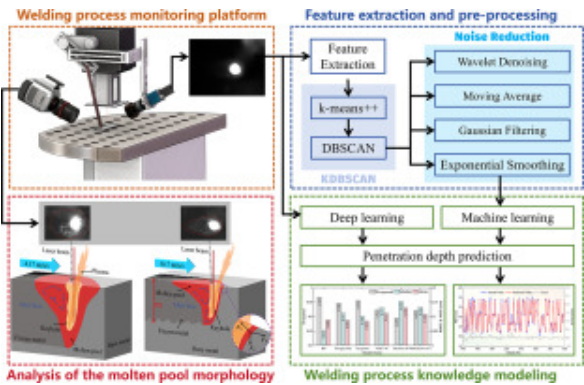
3. Results and discussion

4. Conclusions

CRediT authorship contribution statement

underwent significant changes, the surface areas of keyhole and molten pool exhibited nonlinear decreasing trends. Furthermore, the geometric features of keyhole visual images were extracted, and a processing method of KDBSCAN combined with exponential smoothing was proposed to reduce noise in keyhole features owing to the significant fluctuation of the molten pool caused by the periodicity of laser power. Compared with the original features, the signal-to-noise ratio of the processed features improved 6.82dB on average. In this study, the maximum depth of the weld area within a single pulse period was defined as the penetration depth, and several penetration depth prediction models based on processed keyhole features were established. The results showed that compared with classical regression algorithms, the built multi-layer neural network model had the best performance with R^2 of 0.969. In addition, a variety of deep learning models based on transfer learning presented slightly weaker performance when trained by raw keyhole images. Deep learning model is expected to have a better accuracy of the penetration depth prediction with the support of more training data in the future.

Graphical abstract

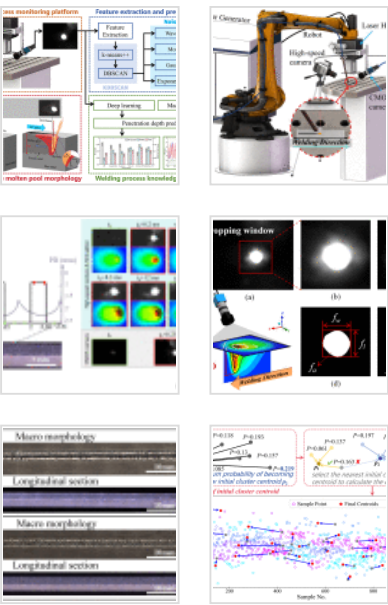


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Keywords

- Declaration of competing interest
- Acknowledgment
- Appendix A. Supplementary data
- References
- Show full outline ▾

Figures (20)



Show 14 more figures ▾

Tables (7)

- Table 1. Main welding process paramete...
- Table 2. Experimental design of variable ...
- Table 3. Standard deviation and sample ...

Pulsed laser welding; Dynamic behavior of molten pool; Keyhole feature extraction;
Machine learning; Deep learning network; Penetration depth prediction

Table 4. The average SNR of feature sign...

Table 5. Evaluation of various PD predict...

Table 6. Number of model parameters a...

[Show 1 more table](#) ▼

[◀ Previous article in this issue](#)

[Next article in this issue ▶](#)

1. Introduction

Titanium alloys have been widely used in aerospace field due to good corrosion resistance, high specific strength and temperature resistance. As a high-efficiency connection technology with the advantages of high precision, low heat input, high power density, etc., pulsed laser welding is commonly used process in manufacturing of titanium alloy rocket grid fin [1]. However, the weld penetration depth (PD) shows a complex fluctuation attributed to the unique cyclic characteristic of laser power during pulsed laser welding, which will affect the mechanical properties of the joint in severe cases [2]. Therefore, an accurate detection of the PD is crucial for determining the joint quality and feedback control of the welding process.

Three-dimensional numerical simulation is a crucial approach to reveal the dynamic behavior of the molten pool [3]. Combining a 3-dimensional multi-physics model with experiments, Jiang et al. [4] systematically investigated the thermo-fluid dynamics and solidification characteristics between continuous-wave and pulsed laser welding processes. It was revealed that pulsed laser welding achieved greater penetration depth with equivalent heat input. Although numerical simulations characterized the keyhole evolution, the computational process was highly dependent on physical modeling parameters. Improper parameters may lead to the distortion of the forming results. Obtaining the actual physical and chemical properties of the molten pool through online sensing measurement, and correcting the parameters of the numerical model according to the measured data is an effective way to improve the accuracy of the simulation results.

Extras (1)

 [Supplementary material](#)

The joint status is usually determined by the electrical [5], acoustic [6], and optical signals [7,8] generated in the welding process. Huang et al. [9] extracted plasma electrical signal features to characterize the laser welding defects and established a prediction model of weld quality with an average prediction accuracy of 90.16%. However, since the electrical signal needs to be measured by inserting an electronic probe into the plasma, the acquisition conditions are relatively harsh. By collecting and analyzing the acoustic signal characteristics of laser welding process, Yusof et al. [10] demonstrated the feasibility of online detection of weld penetration. Nevertheless, the acoustic emission detection method is susceptible to environmental noise. In order to improve the adaptability and stability of the welding process detection, Stadter et al. [11] built an optical coherence tomography (OCT) monitoring platform and proposed a weld PD detection method based on optical radiation signals. Compared to expensive OCT equipment, visual sensing is highly cost effective and has been widely used in the online inspection of welding process [12]. Utilizing a CMOS camera to passively acquire keyhole images, Tang et al. [13] established a weld defect detection system based on the keyhole contour features. Using the active visual acquisition system, Lu et al. [14] determined the weld penetration status accurately by extracting the surface information of the molten pool. It is obvious that visual sensors are equipped to obtain the features of the molten pool that are closely related to the weld quality timely and accurately.

Previous studies have shown that the dynamic behavior of the molten pool is strongly correlated with the penetration state in the continuous laser welding [15]. However, how to characterize the dynamic behavior of the molten pool and establish the quantitative relationship between the surface morphology of molten pool and PD are the key to achieve the prediction of PD. To solve this problem, Zhao et al. [16] selected 10 features of keyhole images based on wrapper feature selection method and achieved the classification prediction of penetration state. The raw image data contains abundant information. Through feature extraction, details that represent the stability of the welding process are obtained. Current studies indicate that these extracted features are useful to identify the penetration states and forming quality [17]. However, satisfactory results in

predicting PD have not yet been obtained. This is because that the extracted features are subject to considerable fluctuations as a result of the complex welding conditions, which negatively affects the accuracy of PD prediction models. Therefore, comprehensive processing of features is crucial for improving the accuracy of PD prediction. He et al. [18] proposed a processing approach that combines the HDBSCAN clustering, percentile filtering and exponential moving average method. Compared to DBSCAN and K-means clustering algorithms, the proposed method significantly enhances the accuracy of PD value prediction, reducing the average error by 45% at best. The study demonstrates the effectiveness of the optimization algorithm based on DBSCAN in noise reduction, while simultaneously highlighting that the algorithms rely heavily on human expertise and experience. In recent years, deep learning algorithm has attracted wide attention in the field of welding process detection due to great computing power and high accuracy [[19], [20], [21]]. Based on the continuous welding data collected by CCD cameras, Li et al. [22] trained a multi-task regression model for predicting PD and penetration width at the same time, which has a better performance than machine learning models such as support vector machine (SVM) and K-nearest neighbor (KNN). By processing images such as convolution and pooling directly, it is more likely to extract effective physical information [23,24]. The melting and solidification of metal are periodic due to the pulsed power output during pulsed laser welding, which lead to the regular fluctuation of PD. Since the real-time prediction of titanium alloy pulsed laser welding PD is difficult, the performance of machine learning model and deep neural network in the prediction of PD under this process needs to be further discussed. Therefore, this study intends to explore the feasibility of using keyhole visual features for the detection of weld PD value and establish a PD prediction model for pulsed laser welding process.

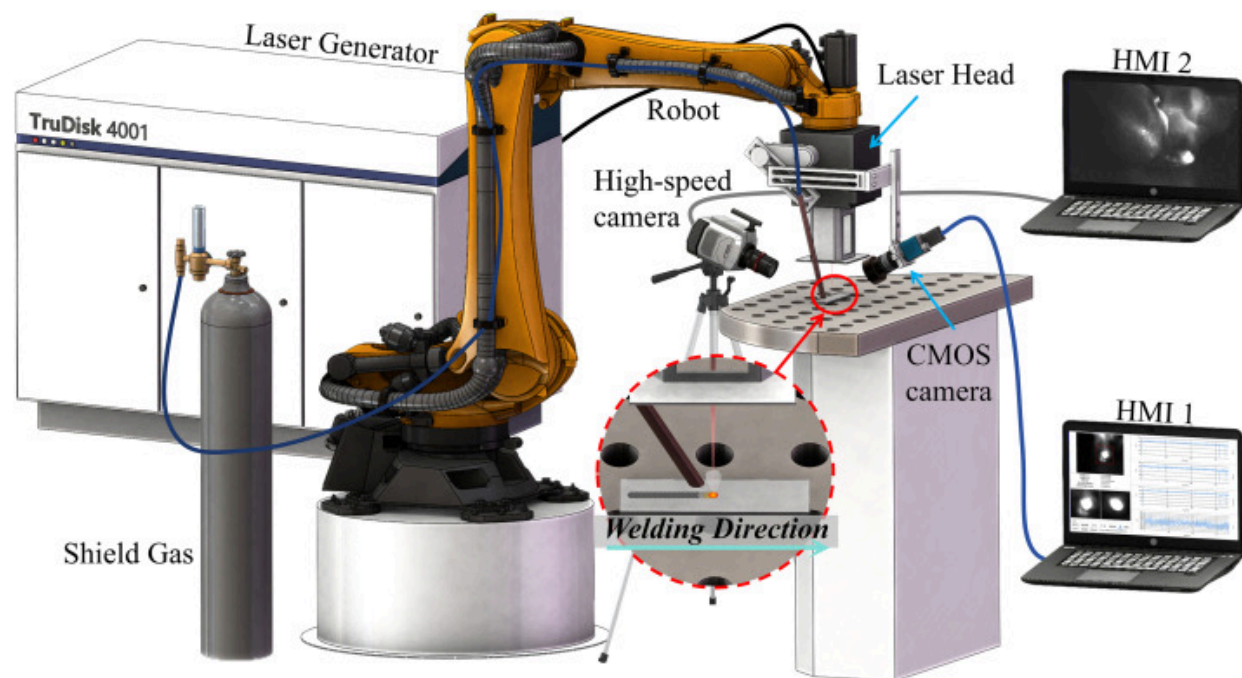
The remainder of this research is divided into five parts. [Section 2.1](#) introduces the laser welding visual signal monitoring platform and designs a series of dynamic welding experiments. In [2.2 Feature extraction](#), [2.3 Outlier detection and noise reduction](#), the clustering analysis combined with the noise reduction algorithm is applied for pre-processing to enhance the quality of the keyhole features. Combining the morphology of

the molten pool acquired by high-speed camera and the 3-dimensional fluid simulation results, [Section 3.1](#) analyzes the influence of process parameters on the surface behavior of the keyhole and the weld PD. Then, a BP neural network model for predicting PD values is constructed in [Section 3.2](#). The prediction performances of machine learning and deep learning models are compared and analyzed. Finally, the main conclusions are summarized in [Section 4](#).

2. Experimental setup and method

2.1. Experimental setup

The experimental platform is shown in [Fig. 1](#). In this research, 150mm×40mm×4mm TC4 titanium alloy is used as the base metal. The PD detection system is mainly composed of a laser welding subsystem and a visual sensing subsystem. The laser welding system includes a laser generator, a laser head and a KUKA six-axis robot. The laser pulse period is set to 33ms according to the production process specification. The laser head is mounted at the end of the robot, cooperating with the robot movement to perform the welding. To better understand the evolution of the temperature field and the molten pool flow process, this study employs ANSYS® FLUENT software to develop a transient 3-dimensional simulation model of laser pulsed welding of TC4 titanium alloy (described detailedly in Supplementary materials). The model is used to elucidate the inherent relationship between keyhole features and PD, by which the influence of welding speed on visual signals and weld quality is demonstrated.



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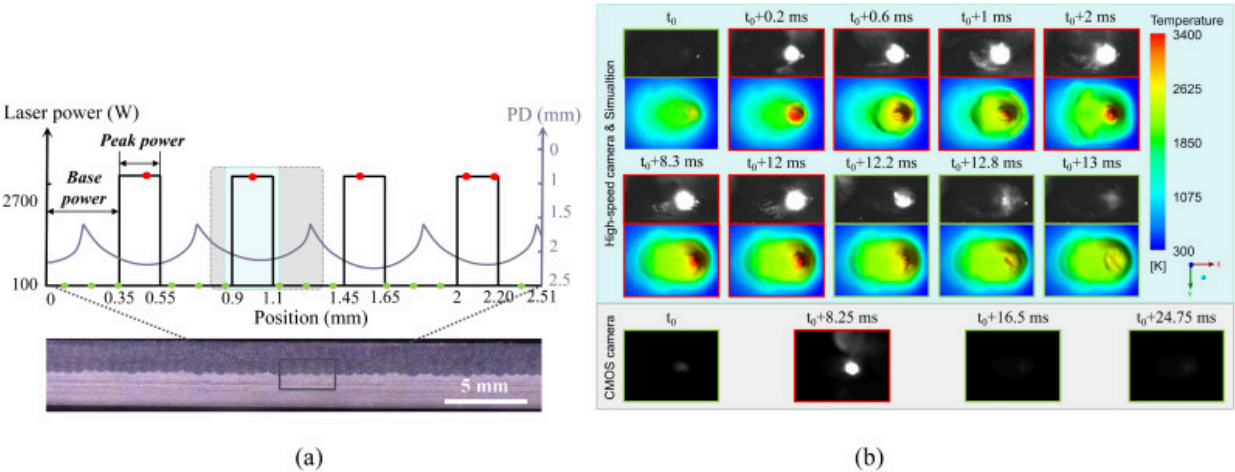
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Fig. 1. Schematic diagram of titanium alloy laser welding experimental platform.

In order to detect the welding process and predict the PD, a visual sensing system is set up. The system consists of a high-speed camera, a CMOS industrial camera and an industrial computer. The high-speed camera in-situ takes picture of the keyhole at a position perpendicular to the welding direction, with an angle of 45° to the horizontal direction. The camera is equipped with an 808nm filter. The parameters are set as follows. The exposure time is set to $8\mu\text{s}$, the aperture is set to F22, and the frame rate is set to 5000 frames/s.

The CMOS camera is fixed to the front side of the robot through clamps and collects images at a 30° angle to the horizontal direction. The camera is equipped with an 808nm filter and a transmittance of 0.3%, and the exposure time is set to $100\mu\text{s}$ with an aperture

of F1.4. The acquired images are transmitted to an industrial computer for real-time feature extraction. In order to acquire visual images at the peak pulse time, the frame rate is set to 121.2 frames/s. It means that 4 images are captured in each pulse cycle, so that at least one keyhole picture could be acquired during the peak pulse time. The correspondence between the acquisition frequency and the pulse cycle is shown in Fig. 2, along with the molten pool surface images captured by the vision system during a single cycle. The keyhole is rapidly formed within 0.2ms after the laser peak comes and disappears within 1 ms after the peak ends. The main experimental platform process parameters are shown in Table 1.



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Fig. 2. Molten pool images and weld longitudinal metallography acquired with typical process parameters in Table 1 (at welding speed of 16.7mm/s). (a) Corresponding diagram of laser pulse period and weld PD. (b) Comparison between the visual images and simulation results.

Table 1. Main welding process parameters.

Parameter	Value
Laser peak power	2700W
Beam parameter product	4mm·mrad
Shield gas flow	30L/min
Laser pulse period	33ms
Laser base power	100W
Laser wave length	1030nm
Shield gas	99.9%Ar
Laser pulse peak period	12ms

Laser power and welding speed are the main parameters of pulsed laser welding process. Since the power regulation authority of the production equipment is not obtained, the welding speed is selected as the moderator in this research. The experiments of varying welding speed are designed and listed in [Table 2](#).

Table 2. Experimental design of variable welding speed.

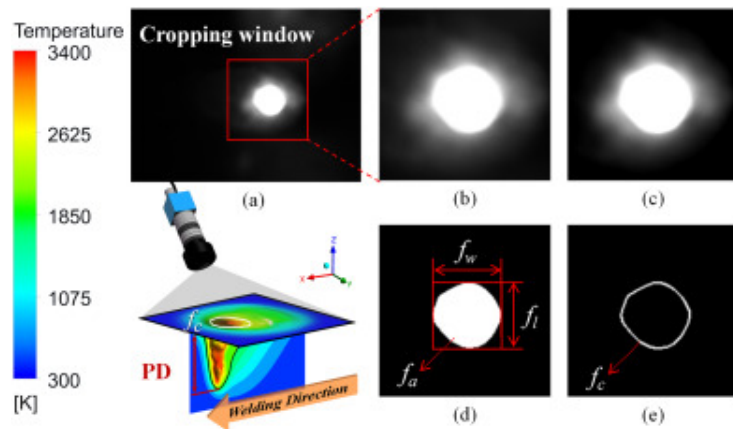
Experiment no.	1	2	3	4
Welding speed/mm·s ⁻¹	4.17	8.35	12.53	16.7

2.2. Feature extraction

During laser welding process, thermal radiation from the inner walls of the keyhole and surrounding material generates photons, which are then captured by the CMOS sensor. As the welding speed increases, the heat input decreases, leading to a reduction in the recoil force of the metal vapor (F_p). Consequently, the surface diameter of keyhole decreases,

which suppresses the outward propagation of light, resulting in a further reduction in captured keyhole surface features.

The keyhole features processing flow chart is shown in Fig. 3. Appropriate region of interest (ROI) is cut from the original images, as shown in Fig. 3(b), so as to reduce redundancy information and improve computing efficiency. Due to the noise and environmental interference caused by intense plasma eruption in the process of signal acquisition, the median filtering method as shown in Fig. 3(c) is adopted for filtering out the noise. Then, the binarization is used to separate the keyhole from the surrounding environment. The length feature f_L , width feature f_W and area feature f_A of the keyhole are obtained by traversing the pixels, as shown in Fig. 3(d). Finally, in Fig. 3(e), the number of pixels in the keyhole circumference feature f_C is obtained with the canny edge detection algorithm.



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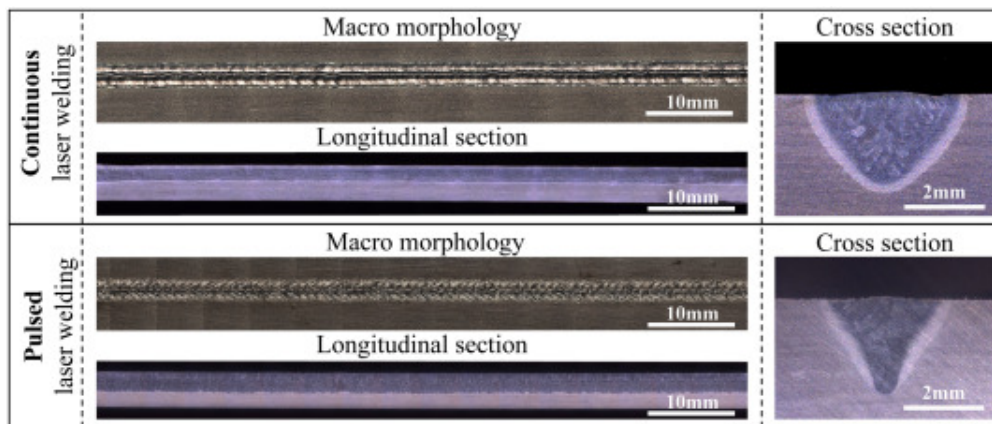
Fig. 3. Keyhole feature processing flow chart. (a) Raw image. (b) Cropped ROI. (c) Median filtering. (d) Binarization. (e) Edge detection.

According to the experimental conditions, the CMOS camera acquires at least one keyhole image during the pulse peak time in a single pulse period. In this study, the image with

the maximum f_A is the target keyhole data of one single period. Therefore, images with the maximum f_A are selected to form a new image set.

2.3. Outlier detection and noise reduction

The keyhole features fluctuated sharply during pulsed laser welding. On the one hand, the collected images of molten pool alternates between light and dark because of the pulsed laser power output. Accordingly, the keyhole features greatly fluctuate. On the other hand, the pulse overlap rate decreases with the increase of welding speed. The heat effect of adjacent pulses on the molten pool is reduced, and the longitudinal section of PD gradually changes into a wavy shape as shown in Fig. 4 [25]. This power output mode makes the visual information of weld surface change periodically, which greatly reduces the stability of features of images. Therefore, this study extracts keyhole features from visual images, and combines KDBSCAN algorithm with exponential smoothing (ES) to preprocess the features.



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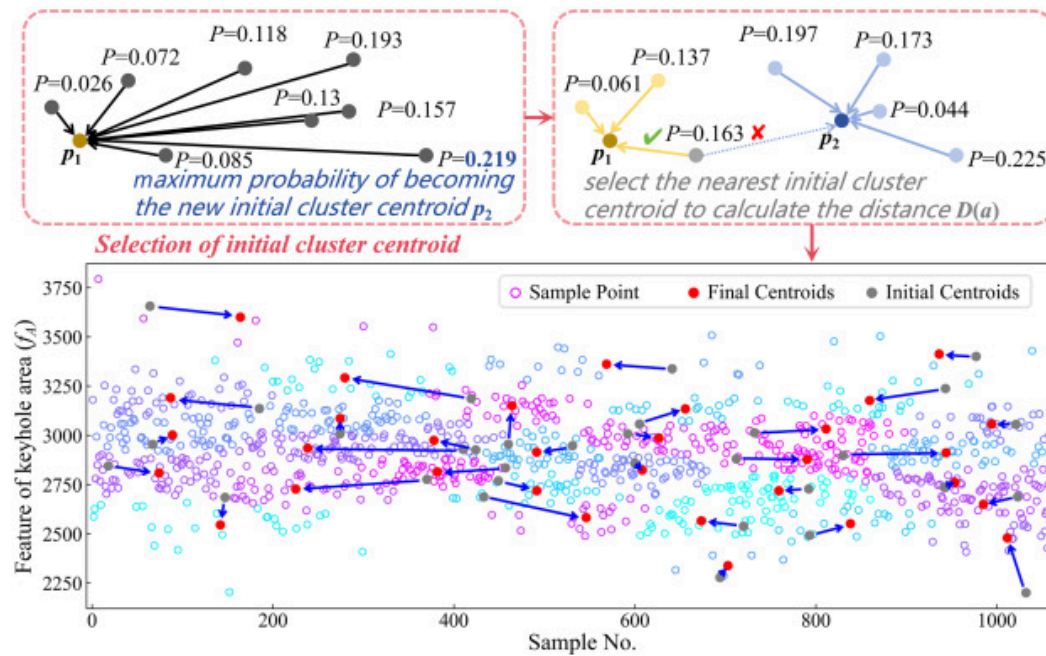
Fig. 4. Comparison between continuous welding and pulse welding of titanium alloy.

Density-based spatial clustering of applications with noise (DBSCAN) is a clustering algorithm which has been widely used in outlier detection. Based on this algorithm, Gholizadeh et al. [26] developed k-means based DBSCAN (KDBSCAN) in 2021, which was able to process large amounts of data efficiently. The basic process is as follows:

- (1) The k-means++ algorithm is applied to divide the data set. The first cluster centroid p_1 is randomly selected in the global space. For every non-centered data $a \in X$, calculate the Euclidean distance $D(a)$ between a and the nearest cluster centroid to a . Calculate the probability $P(a)$ that the non-centered point a is the new cluster centroid:

$$P(a) = \frac{D(a)^2}{\sum_{x \in X} D(x)^2} \quad (1)$$

where X is the set of non-centered points. According to the obtained probability distribution, the new cluster centroids are iterated by random selection until all cluster centroids are selected. Then, the mass center of each cluster is calculated as the final centroid of that cluster. Taking the f_A acquired under the welding speed of 4.17 mm/s as an example, the result of k-means++ clustering and cluster centroids optimization directions are shown in Fig. 5.



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Fig. 5. Division of feature data based on k-means++ clustering.

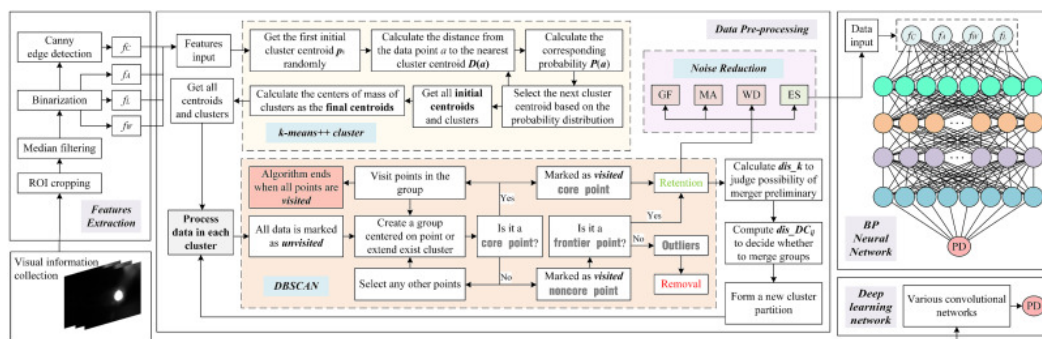
- (2) Apply the DBSCAN algorithm to each cluster. All points are marked as “unvisited”. Starting from any data point s , it is marked as a core point if there are at least $s_{\min}$ data in the ε neighborhood of s , otherwise it is marked as a non-core point and a new point is selected. When s is the core point, s is marked as “visited”. A group centered on s is created, and all other points in the neighborhood within radius ε are checked to see if they are core points. If so, the group is extended until it cannot be extended and then look for the rest points that is “unvisited”. The algorithm ends when all points have been visited.

- (3) The algorithm evaluates the potential overlap or adjacency between clusters by calculating the distance dis_k between their centroids. If dis_k is less than or equal to ε , the clusters are considered similar and proceed to further inspection. At this stage, the distance between data points within the closest DBSCAN group in the two clusters (dis_D_{ij}) is computed to verify boundary contact or overlap. If dis_D_{ij} satisfies formula (2), they are merged into one cluster.

$$dis_D_{ij} = dis_DC_{ij} - (dis_{ri} + dis_{rj}) \leq \varepsilon \quad (2)$$

where dis_DC_{ij} denotes the distance between the centers of the two DBSCAN groups in the two k-means++ clusters; dis_{ri} and dis_{rj} are the distances between the center of the i and j groups and the point farthest from them in the same groups, respectively.

- (4) After completing the merge marking of clusters, the algorithm applies DBSCAN clustering algorithm to each merged cluster again. This process helps to effectively identify and deal with outliers when there is slight overlap or noise between clusters. The processing process based on keyhole features is shown in Fig. 6.



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Fig. 6. Processing schematic of keyhole features and PD prediction.

It has been proved that the number of cluster centers k is negatively correlated with the execution time of the algorithm [27], which means that k not only dictates the grouping strategy, but also affects the computational complexity within each cluster. In order to balance the partition speed of clusters and the efficiency of intra-cluster computation, the square root of the number of samples n is considered as k .

The KDBSCAN algorithm relies on empirical judgment and manual trial-and-error when determining the ε and s_{min} . This process is random, making it challenging to achieve consistent clustering results across different feature data. Starczewski et al. [28] proposed a DBSCAN automatic parameter finding method: $D(a)$ was sorted by value to obtain a distance decline curve, and ε was determined by analyzing the region with sharp changes in the curve and the related distance changes. At the same time, s_{min} is determined according to the distance relationship between points. This method is applied to each cluster after k-means++ grouping, and the mean value of all group ε and the mode of s_{min} are obtained as the parameters of DBSCAN.

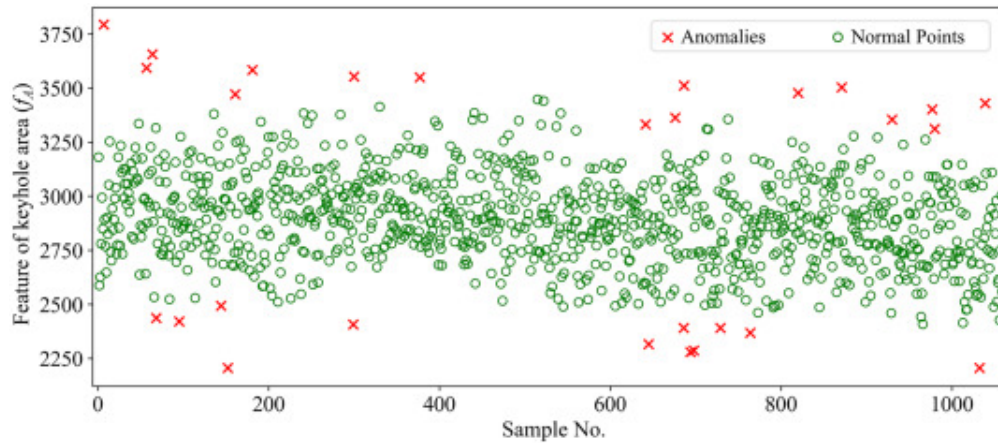
The initial grouping of target data through k-means++ in advance reduces the computational complexity, which significantly improves the pre-processing efficiency and provides a possibility for real-time prediction of PD [29]. Furthermore, the DBSCAN algorithm requires manual trial and error to determine its parameters, whereas the KDBSCAN algorithm automatically determines the parameters, thus avoiding the clustering performance issues caused by subjective parameter selection and effectively improving the adaptability of KDBSCAN to density changes.

In this paper, outlier detection algorithms commonly used in statistics are applied to process the same set of feature data, such as interquartile range (IQR), methods based on local outlier factor (LOF) and percentile division. Standard deviation was used to evaluate the treatment effect, as shown in Table 3. It is seen that the feature data show different degrees of optimization effect after processing by various outlier detection methods,

among which LOF and KDBSCAN methods perform better. However, the LOF algorithm improves the data quality at the cost of removing more samples, which causes the feature data to lose some valid data and distort to a certain extent. The combination of k-means++ and DBSCAN can accurately identify outliers and retain the maximum amount of data to generate more accurate clustering results. Fig. 7 shows the outliers of f_A obtained by KDBSCAN. In the subsequent processing, these outliers will be deleted in order to obtain the keyhole features that can accurately predict the PD.

Table 3. Standard deviation and sample number of f_A after removing outliers by different algorithms.

Algorithm	Standard deviation	Sample number after removing outliers
Raw data without processing	217.109	1059
IQR	204.894	1001
LOF	190.477	853
Percentage division	199.942	1007
KDBSCAN	192.477	1008



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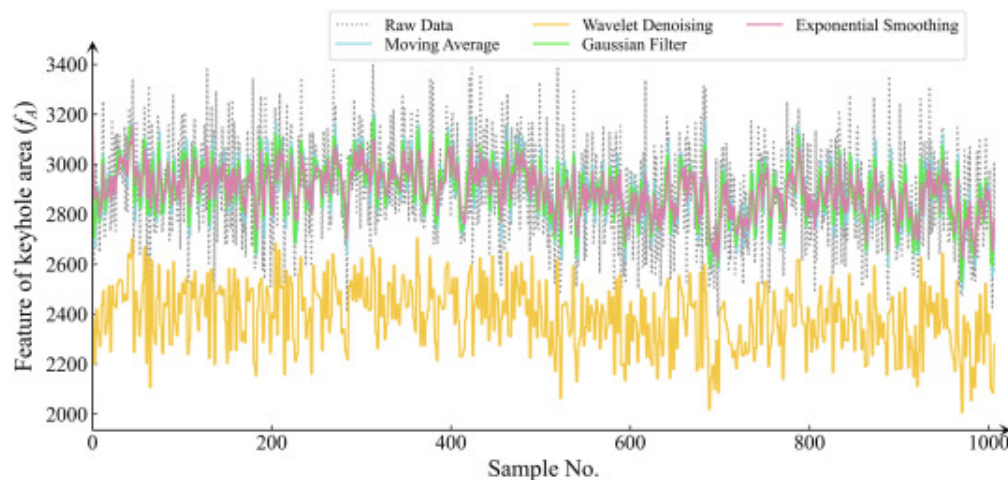
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Fig. 7. KDBSCAN-based outlier labeling for feature data.

ES is a method for time series data that predicts future values by assigning different weights to historical data. The smoothing value of sample is the corresponding denoising value of the original sample. For a given series $B=\{b_1, b_2, \dots, b_n\}$, calculate the smoothing value S_t of the data as follows:

$$S_t = \alpha b_t + (1 - \alpha)S_{t-1} \quad (3)$$

where t is the time, $t \in [1, n]$, $t \in \mathbb{N}$. S_t is the smoothing value of the data at instant t , b_t is the original value of the data at instant t , and α is the smoothing factor that determines the weight of the most recent data. ES has the characteristics of weight decline. The latest data points have the largest weight, and the weight of previous data points gradually decreases, so that the method respond to change trend in new data quickly. This study compares the processing effects of various noise reduction algorithms on keyhole features at different welding speed, including moving average (MA), wavelet denoising (WD) and Gaussian filtering (GF). The processing effect of f_A at 4.17 mm/s is shown in Fig. 8.



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Fig. 8. Comparison of processing effects of different noise reduction methods.

The signal-to-noise ratio (SNR) is used as the evaluation indicator of signal noise reduction effect:

$$SNR = 10 \cdot \log_{10} \left(\frac{SignalPower}{NoisePower} \right) \quad (4)$$

where *Signal Power* and *Noise Power* are obtained from (5), (6) respectively:

$$SignalPower = \frac{1}{N} \sum_{i=1}^N x_i^2 \quad (5)$$

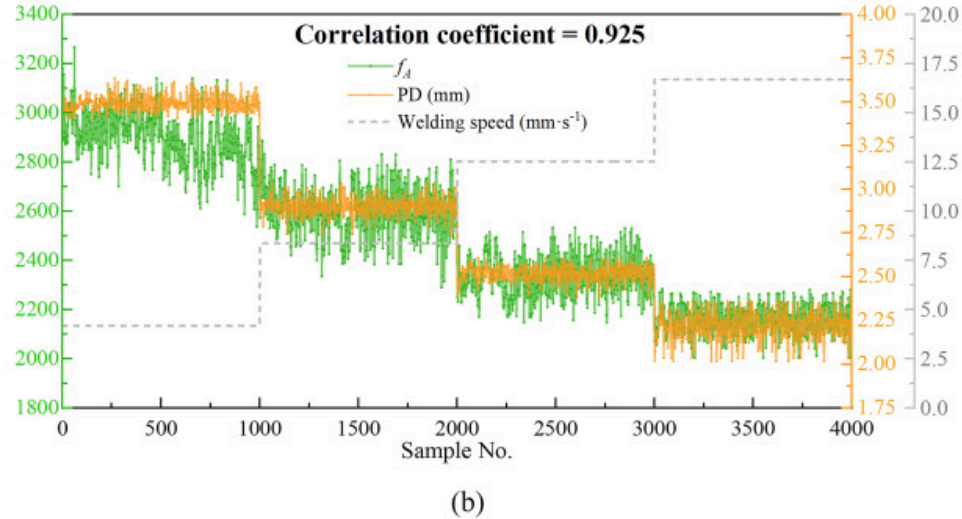
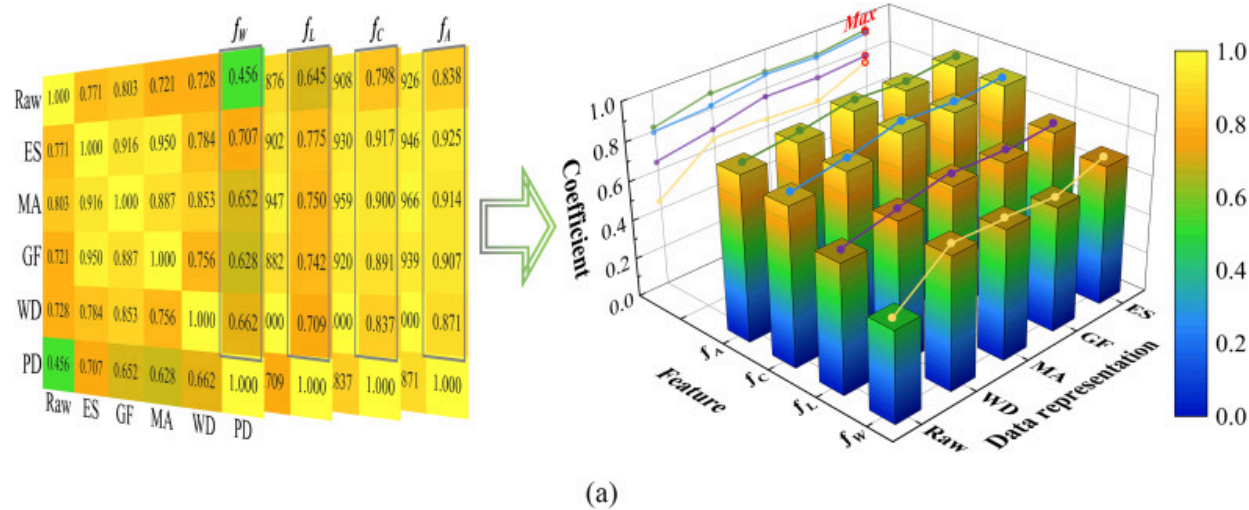
$$NoisePower = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \quad (6)$$

where N is the number of signal samples, x_i represents each sample value of the signal, and μ represents the mean value of the signal. High SNR means high quality of signal and strong anti-interference ability. The average SNR of keyhole features processed by the

above algorithms at different welding speeds is listed in Table 4. It is seen that signals processed by ES obtain the highest SNR. It is obvious from Fig. 8 that WD is difficult to truly reflect the original value of data. MA and GF have little difference in processing effect. GF has high computational complexity when the amount of data is large. MA is not sensitive to the new data because it gives the same weight to all data points within a window range. ES reduces the variance of the data deviating from the mean significantly. Furthermore, Spearman correlation coefficient is used to analyze the relationship between denoised feature data and PD. The results are shown in Fig. 9(a). Obviously, the keyhole features that are denoised by the ES have the highest correlation coefficient with PD. The feature f_A , which shows strong correlation with PD, is selected to construct the relationship curve among it, PD and welding speed, as depicted in Fig. 9(b). Therefore, KDBSCAN algorithm combined with ES is suggested to be the best denoising method to pre-process keyhole features.

Table 4. The average SNR of feature signals processed by different noise reduction methods.

Feature	Raw data/dB	ES/dB	GF/dB	MA/dB	WD/dB
f_W	22.68	29.50	27.93	27.26	24.03
f_L	21.37	28.63	26.83	26.13	22.55
f_C	21.46	27.57	26.22	25.59	22.20
f_A	20.98	28.07	26.11	25.46	21.62



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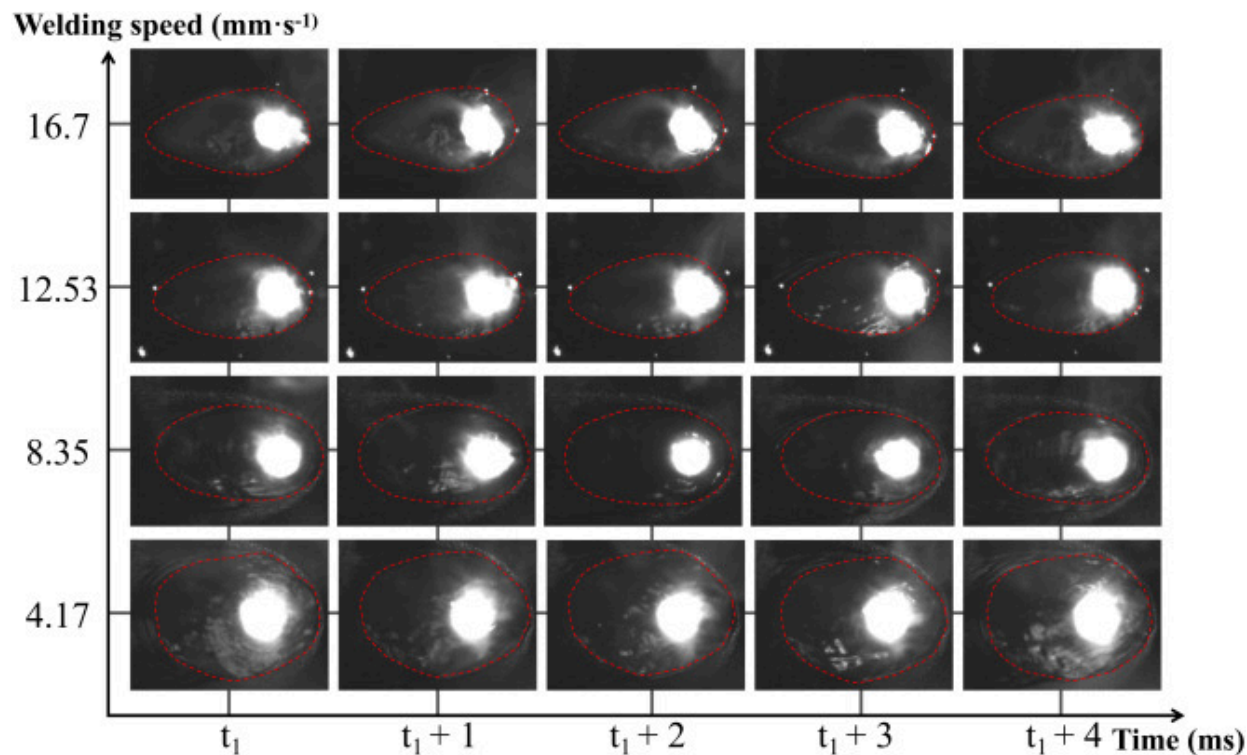
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Fig. 9. Relationship between denoised keyhole features and PD. (a) Correlation analysis of denoised keyhole features and PD. (b) Curves of KDBSCAN - ES processed f_A , welding speed and PD.

3. Results and discussion

3.1. Effect of welding speed on the molten pool morphology

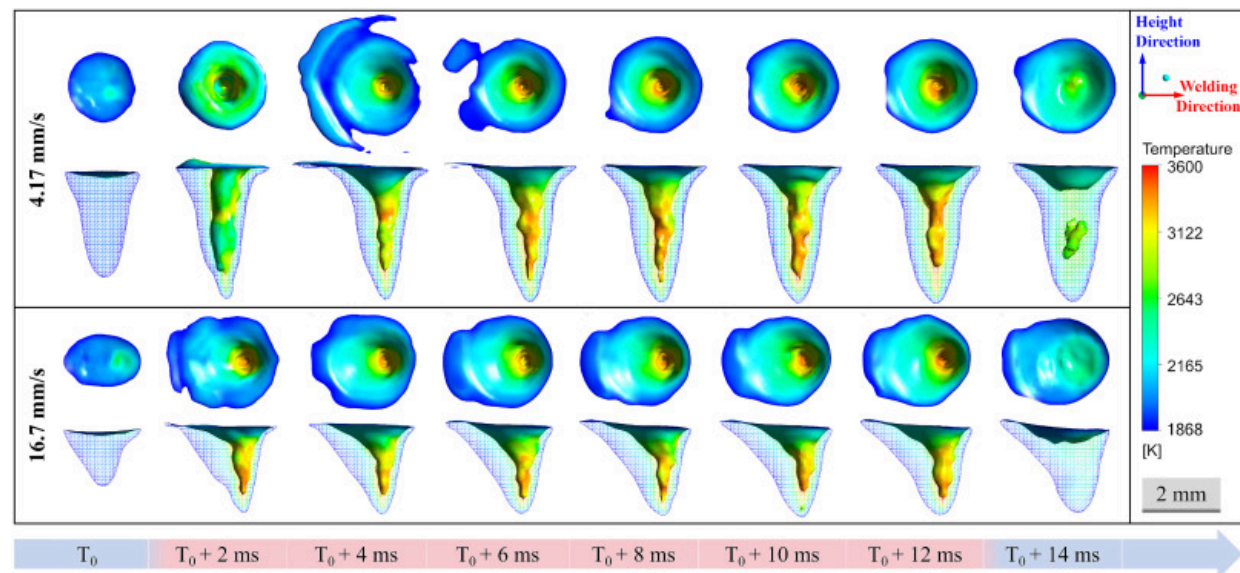
The surface morphology of titanium alloy pulsed laser welding pool acquired by the high-speed camera under different welding speeds is shown in Figs. 10, and Fig. 11 compares the dynamic evolution of the weld pool and keyhole at the peak laser pulse when the welding speed is 4.17 mm/s and 16.7 mm/s, respectively. The top view in Fig. 11 depicts the liquid-gas interface at temperatures exceeding the melting point of Ti6Al4V (1868K), while the front view presents the outer contour of the keyhole's inner wall. The grid in the front view represents the temperature field of the longitudinal section at the weld center surpassing the melting point. In this study, the boundary of the grid is approximated as the molten pool profile in the longitudinal section.



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Fig. 10. Surface morphology of titanium alloy pulsed laser welding pool at different welding speeds.



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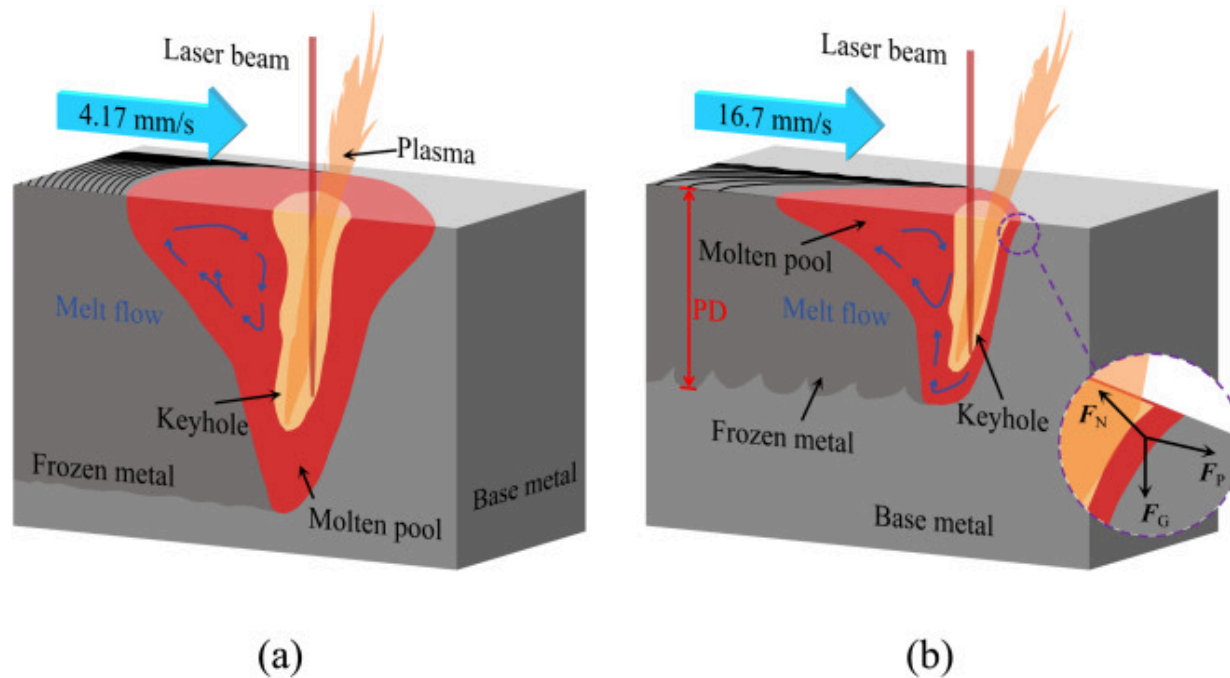
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Fig. 11. Dynamic evolution process of pulsed laser welding at different welding speeds.

At the welding speed of 4.17 mm/s, the welding heat input is large, and the welding mode is deep penetration welding. The weld forming is closer to that of continuous laser welding. As depicted in Fig. 10, compared to other welding speeds, the weld pool surface area at this welding speed is significantly larger. Combined with the high-speed connected acquisition data and simulation results, it is seen that the keyhole is centered in the molten pool to the side of the welding direction, the heat sources in front and back of the molten pool are relatively symmetrical. Due to the high overall heat input, a large configuration of molten pool has formed before the laser reaches its peak power. The laser acts on the metal of the keyhole wall, triggering intense evaporation, forcing the liquid around the keyhole to advance towards the front of the molten pool or flows

through the sides of the pool towards the end side. It is significant that the laser-heated liquid metal advances most towards the rear of the molten pool, resulting in significant liquid fluctuations. This instability is associated with the rapid formation of the molten pool. In larger molten pools, these fluctuations are low-frequency and persistent. It lasts for up to 6ms before gradually diminishing, ultimately forming a low-eccentricity elliptical pool on the surface.

When the welding speed is increased to 8.35mm/s, the molten pool area starts to decrease and the front side of the molten pool remains smoothly semicircular. The liquid metal that expands towards the front side of the molten pool also starts to decrease, and the contour of the front side of the keyhole is closer to the boundary of the molten pool. It is seen that the welding speed has a significant effect on the morphology of the molten pool and the keyhole, due to the fact that the welding speed directly affects the heat input into the molten pool. When other process parameters remain constant, as the welding speed increases, the laser action time on the weld and heat input per unit volume decrease. As depicted in [Fig. 12](#), the effective PD of pulse laser welding is defined as the distance from the surface of the workpiece to the deepest position of the fusion metal with every single pulse [30]. The decrease of heat input results in the decrease of the molten pool volume, which is reflected in the monotonic decrease of the width and depth of the molten pool. At the same time, the cooling rate increase with the increase of welding speed, which means that there is not enough time for heat to be evenly distributed to the surrounding metal. The back end of the molten pool cools too fast for the metal to flow before it solidifies, resulting in an irregular, sharp shape, known as a “tailing”.



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Fig. 12. Mechanism diagram of laser-induced molten pool. (a) Welding speed of 4.17 mm/s. (b) Welding speed of 16.7 mm/s.

When the welding speed increases to 12.53 mm/s, the heat input continues to decrease. The solidification of the liquid metal is accelerated, and the tailing of the molten pool becomes obvious. During the action of the laser beam with the workpiece, the laser acts directly on the deep layers of the material through the keyhole, and the heat is mainly transferred through the side wall of the keyhole. It has been found that the increase in speed causes a significant narrowing of the depth direction of the keyhole, which makes it difficult for the heat to be transferred directly to the workpiece through the inner wall of the keyhole, resulting in a decrease in the molten pool volume. In addition, it is seen from Fig. 10 that the surface fluctuation of the liquid metal is more intense. This is because the fluctuation of the molten pool is directly affected by the variation of the

keyhole morphology. When the keyhole is stable, the molten pool usually behaves smoothly. As is shown in Fig. 12, when the balance among the gravitational force of the molten liquid metal F_G , the pressure of the liquid metal on the unmelted inner wall at the front side F_N and F_P is disturbed, the inner wall of the keyhole fluctuates slightly. The dynamic behavior of the keyhole causes fluctuations in the liquid, and the frequency of the fluctuations of the weld increases with the welding speed.

When the welding speed increases to 16.7mm/s, the length of the molten pool increases further while the width and the surface area of the molten pool are further reduced. The geometric outline is close to a cone. It is concluded from the simulation results that when the keyhole depth is significantly smaller than that at a welding speed of 4mm/s, the heat transfer in the depth direction is greatly reduced. The heat at the rear side of the molten pool extends in a long “tailing”, which is consistent with the behavior of the surface morphology of the molten pool. The apparent changes in the molten pool and keyhole are closely related to the physical properties of titanium alloys, such as thermal conductivity and melting point. Titanium alloys have a high melting point of 1868K and a thermal conductivity of 7.955W/m·K, which is only 1/10 that of aluminum alloys and 1/5 that of iron. These properties cause titanium alloys to exhibit unique heat transfer mechanism during laser welding. Due to their low thermal conductivity, titanium alloys also have a relatively low thermal absorption rate, making it more difficult for the molten pool to form and maintain.

The low thermal conductivity and thermal absorption rate of titanium alloys result in a large temperature gradient within the molten pool under laser irradiation. This gradient influences the morphology of the molten pool and the features of the keyhole. In summary, the morphology characteristics of the weld pool changes monotonously with the welding speed, indicating a direct correspondence between the surface features of the weld pool and the PD. These variations provide distinct visual features for machine learning models, suggesting that the geometric features of the molten pool can reflect the PD of the weld. Given that high-speed cameras cannot meet the requirements for real-

time detection, this study explores a method for real-time PD detection based on keyhole features captured by a CMOS camera.

3.2. Prediction of penetration depth

3.2.1. Prediction models based on machine learning

In order to predict the PD value of pulsed laser welding, a back-propagation (BP) neural network containing four hidden layers is established. The feature information of the stable stage in the welding process is selected as the input. The total number of samples n is 4000, and the division ratio of training dataset and test dataset is 4:1. The structure of the PD prediction model based on visual images in this study is shown in Fig. 13. In this model, the input includes 4 neural nodes. The hidden layers have 8, 16, 32 and 8 nodes respectively, and the output is the predicted PD. Furthermore, Rectified Linear Unit (ReLU) is selected as the activation function of the first and last hidden layers, and that of the rest hidden layers are hyperbolic tangent functions (tanh). Then, R^2 (R-squared), RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) are used to evaluate the precision of the model.

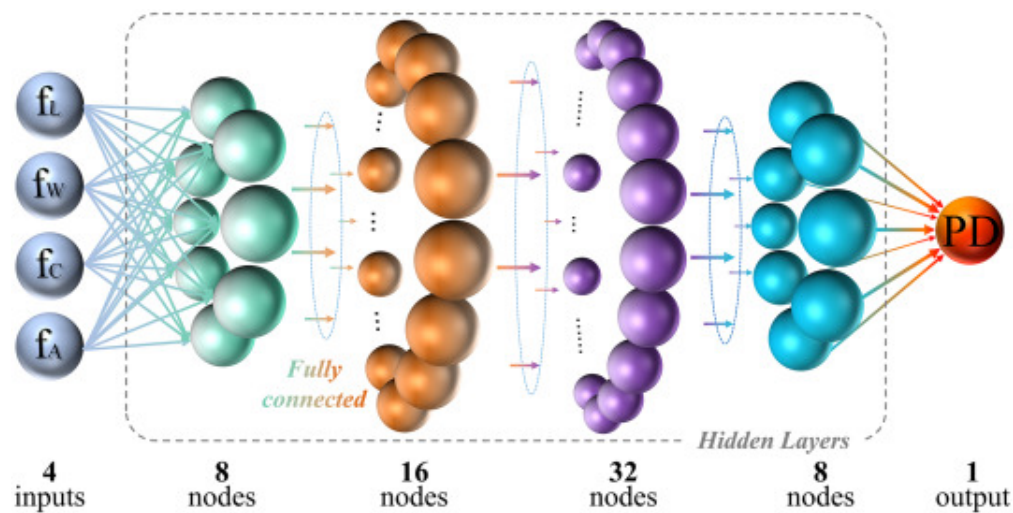
$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (\bar{y}_i - \hat{y}_i)^2} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (9)$$

where $\hat{y}_i$ is the predicted value of the i -th sample, y_i is the actual value of the i -th sample, $\bar{y}_i$ is the sample mean, and n is the total number. The process of machine learning can be regarded as the parameter optimization process of neural nodes based on

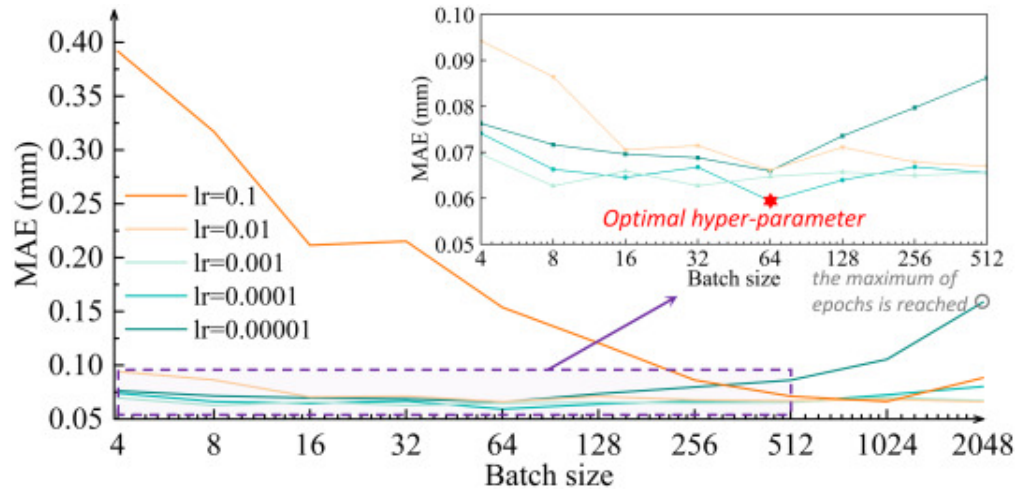
dataset. The key is to obtain the best model architecture by selecting appropriate functions and adjusting its hyper-parameters depending on gradient descent method. The most important hyper-parameters include epoch, batch size and learning rate (lr) [31]. In order to minimize the loss function on the premise that the model is fully trained, a large value of 5000 is set as the number of epochs and an early stopping strategy is adopted. The training will be stopped when the minimum value of loss is not updated after 50 epochs. The batch size defines the sample number applied in each gradient update. The computational efficiency and final performance of the model may be affected significantly if the value is too small or too large [32]. The lr determines the step size of gradient descent, and high lr leads to missing the optimal solution. In this paper, 5 lr s are paired with 10 batch sizes to form 50 combinations, and the optimal hyper-parameter combinations are calculated for the processed feature dataset. The comparison of model performance under different hyper-parameter combinations is presented in Fig. 14.



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Fig. 13. Structure of BP neural network.



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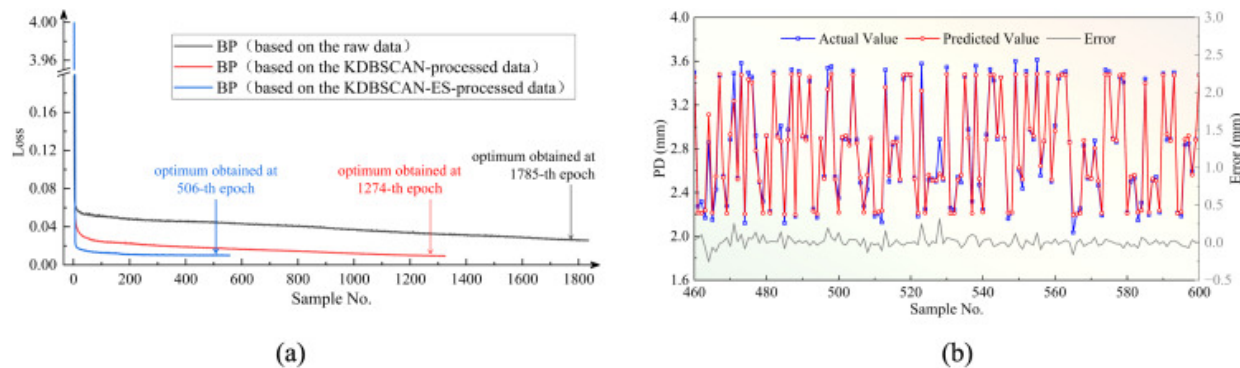
Fig. 14. Structure of BP neural network.

It is observed that MAE decreases and then increases with the increase of lr . MAE always performs poorly when the lr reaches 0.1, which verifies that the larger lr mentioned above is difficult to converge near the optimal solution. Optimal performance of lr is typically observed at $[0.00001, 0.001]$, but even with a sufficiently large number of epochs, excessively small lrs may cause the training process to terminate prematurely, preventing the model from reaching the optimal solution.

Except for a generally large error at a lr of 0.1, MAE also exhibits a trend of initially decreasing and then increasing with the rise of batch size. This is because when the batch size is set too small, the number of samples used for parameter updates within a single epoch is insufficient, while the frequency of updates is excessively high. Consequently, this leads to increased computational overhead and training instability, making it challenging for the model parameters to converge towards the optimal solution. As the batch size increases, the direction of descent becomes more stable, resulting in reduced variation in training loss. However, when the batch size becomes too large, the loss

function may converge to a sharp minima, which could potentially harm the generalization ability of BP model [33]. lr is less disturbed by other parameters when it is 0.001, therefore a combination of a batch size of 32 and a lr of 0.001 is adopted to train the BP network. It is worth mentioning that the optimal hyper-parameter configuration for a given model vary depending on the specific characteristics of the dataset.

Fig. 15 illustrates the prediction results of the model for the test set. The optimal parameters were obtained in the 31st iteration, as shown in Fig. 15(a). As can be seen from Fig. 15(b), the precision of the prediction model for the test set is relatively high, and the error value is basically within the range $[-0.2, 0.2]$, the average error is 0.0126mm. Obviously, a series of pre-processing methods greatly improve the quality of feature data and improve the precision of the model. This is because the dynamic changes of laser plasma plumes and optical noise affect the stable acquisition of visual signals, and the removal of outliers is helpful to reduce the interference of abnormal eruption of plasma plumes on signal acquisition. In addition, data pre-processing improves the generalization ability of the model, making it more robust when dealing with new data in the welding process.



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Fig. 15. Training and testing of BP model based on different data inputs. (a) Visualization of training process. (b) Prediction value and actual value.

Three machine learning models, radial basis function (RBF) kernel-based support vector regression (SVR), linear kernel-based SVR and random forest (RF) regression, are set up as comparisons. As a representative ensemble learning method, RF effectively avoids the overfitting issue inherent in single decision tree model by training multiple decision trees and aggregating their predictions. Furthermore, RF exhibits strong robustness to noise and performs well when confronted with complex and variable input data. The exceptional generalization ability enables model to obtain reliable prediction results across a wide range of regression tasks. SVR addresses complex nonlinear regression problems by implicitly mapping data through the use of different kernel functions. This approach is particularly effective when there is a highly nonlinear relationship between extracted keyhole features and laser welding parameters. SVR minimizes regression errors and provides accurate predictions, demonstrating strong efficiency and accuracy, especially with the appropriate selection of kernels for high-dimensional and nonlinear data. The model evaluation is shown in [Table 5](#).

Table 5. Evaluation of various PD prediction models of pulsed laser welding.

Model name	R ²	MAE	RMSE
BP(based on the raw data)	0.760	0.156	0.234
BP(based on the KDBSCAN-processed data)	0.885	0.109	0.162
BP(based on the KDBSCAN-ES-processed data)	0.969	0.051	0.088
SVR(RBF kernel)	0.922	0.103	0.134
SVR(linear kernel)	0.908	0.116	0.144
RF	0.932	0.083	0.129

Among them, BP neural network has the best performance, with the R² of 0.969, MAE of 0.051, RMSE of 0.088, indicating that the model is capable of achieving accurate prediction of PD. Moreover, the feature data obtained from three different processing

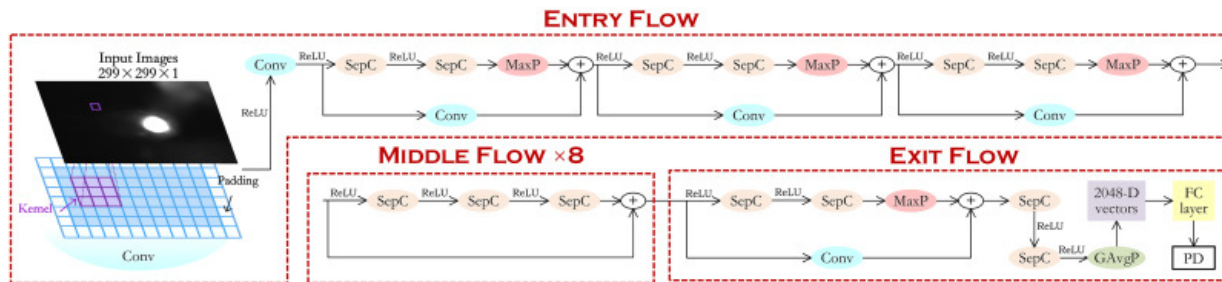
stages as inputs are compared for training the BP model. The results show that both outlier removal and noise reduction contribute to optimizing the data to some extent, thereby confirming the reliability of the algorithm. It is noted that the unique structure of the BP network, along with the carefully selected activation functions, makes the model particularly well-suited for the input of visual features from laser welding keyholes. Additionally, the BP network offers a wide range of hyper-parameters for tuning, allowing the model to be optimized to its performance limits.

3.2.2. Prediction models based on deep learning

With the development of artificial intelligence algorithms, deep learning models have shown great potential and practical application value in the industrial field. Classical models, such as Xception, GoogLeNet and their variants, have achieved excellent results in specific application scenarios, which show strong adaptability even in complex data environments [34]. The middle layer of Xception is a combination of traditional convolution layer (Conv), depthwise separable convolution layer (SepC), max pooling layer (MaxP), global average pooling layer (GAvgP) and fully connected layer (FC). Data are batch normalized after going through all Conv and SepC layers. The model is divided into three parts: Entry Flow, Middle Flow and Exit Flow. Middle Flow is repeated 8 times after Entry Flow.

The structure schematic of Xception for regression prediction is shown in Fig. 16. The originally collected images were resized to dimensions suitable for input into the Xception model and then fed into the model for training. The output feature map of the Entry Flow comprises 4000 feature images of 19×19 pixels. The final output feature map after going through Middle Flow is in the same data format as well. The visualization analysis of layers in Xception is shown in Fig. 17. In feature maps, highlights and color gradients are the focus of model learning. The Entry Flow is the first few layers of the network, which is usually responsible for processing the input data and extracting low-level to intermediate features. In the output feature map of Entry Flow, positions of keyhole centers and contours are extracted as important information, which proves that

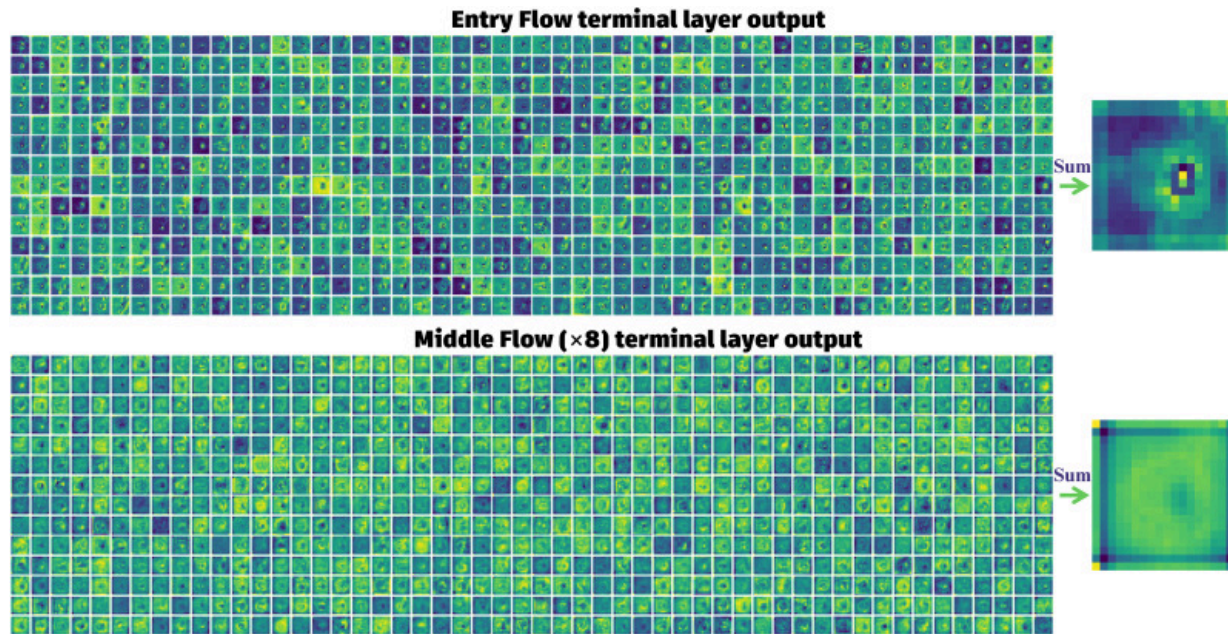
keyhole feature is the key to PD prediction. In the Middle Flow output feature map, the model has shifted to higher-level, more abstract information, including information such as the plasma plume, light reflected from fluctuations in the surface of the molten pool and so on. Therefore, Xception can comprehensively analyze the visual signal of pulsed laser welding, so as to extract enough effective features for PD prediction. The error between the predicted value and the actual value tested by Xception varies in a small range, as shown in Fig. 18, indicating that Xception has good generalization ability.



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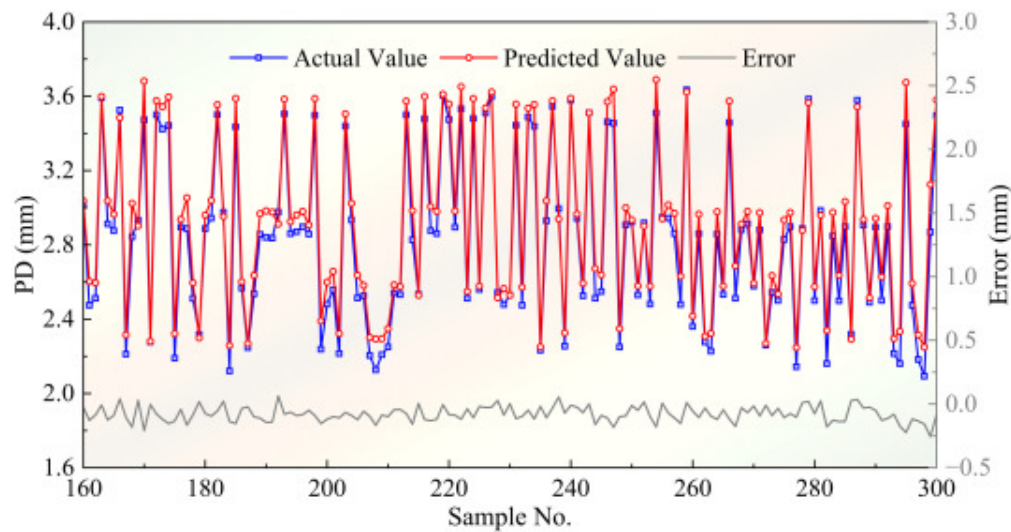
Fig. 16. Structure schematic of Xception.



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Fig. 17. Visualization analysis of layers in Xception.



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Fig. 18. Testing results of Xception model.

In this study, four deep learning models including GoogLeNet, VGG-16, ResNet-101 and MobileNet-V2 are applied to perform regression prediction on the keyhole features by acquired original image dataset. Lightweight network architectures such as ResNet-50, GoogLeNet, and MobileNet V2, reduce computational complexity by incorporating residual structures or Inception modules while maintaining training stability and accuracy. This makes these models particularly well-suited for complex and dynamic conditions encountered in pulsed laser welding. Additionally, the classic VGG-16, which is similar to the Xception, adopts standard convolutional layers and can serve as a reference model for high-precision, complex computations.

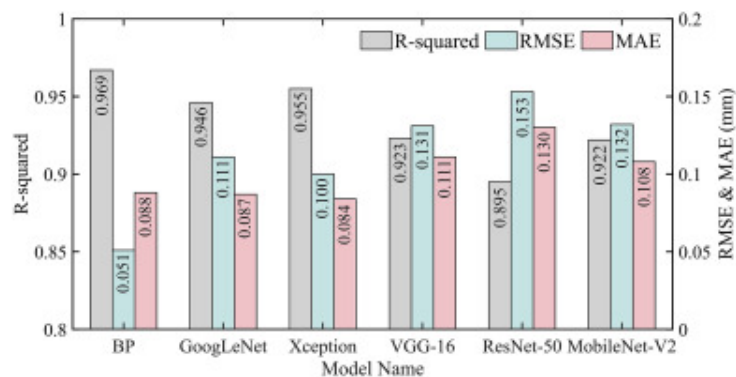
The machine learning and deep neural network tests are performed on the GPU. Due to the different input data formats, the predicted speeds of machine learning models and deep neural networks exhibit a significant difference in magnitude. The deep learning

models take raw images as input, resulting in slower speeds. The predicted speeds on the test set images are shown in the [Table 6](#).

Table 6. Number of model parameters and prediction speed of test image dataset.

Model name	Parameters	Prediction speed on test dataset (frames/s)
Xception	22.9M	68.67
VGG-16	138.3M	57.50
ResNet-50	25.5M	138.02
GoogLeNet	6.9M	331.67
MobileNet V2	3.5M	199.38

The evaluation results of different models are shown in [Fig. 19](#). It is seen that all evaluation indicators of Xception are optimal in the convolutional network model, R^2 is 0.955, RMSE is 0.100, and MAE is 0.084. The results show that the performance of the model is satisfactory with the input of raw image data. The experimental results show that different models adopt different strategies when processing datasets, which means that considering diversified processing algorithms improves the overall performance in the process of model selection and combination significantly.



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Fig. 19. Comparison of BP model and various deep learning models based on three evaluation indexes.

Models on PD prediction during the laser welding process in other studies are listed in [Table 7](#). It is worth mentioning that R^2 is a unitless standardized metric that is free from influence of variations in the original data, effectively reflecting the goodness of fit of a model. R^2 is often regarded as the primary criterion for the reason that it provides a consistent standard for assessment in the evaluation of models based on different input datasets. The research work of the cited references in [Table 7](#) is all carried out based on the continuous laser welding platform. After comparison, it is found that for the nonlinear and complex pulsed laser welding, the BP model can still achieve good penetration depth prediction accuracy, and the prediction results have strong stability under different process parameters, reflecting the effectiveness and reliability of the model.

Table 7. Comparison of regression PD prediction models in laser welding.

Ref.	Acquisition tech.	Welding process	Model	RMSE (mm)	MAE (mm)	R ²
She [35] et al.	CMOS camera	Continuous laser welding	BP	0.019	0.015	0.897
Li [22] et al.	CCD camera		Multi-task CNN	0.082	–	–
Kang [36] et al.	Spectrometer		2D-CNN	–	0.054	0.967
This study	CMOS camera	Pulsed laser welding	BP	0.088	0.051	0.969
			Xception	0.084	0.100	0.955

4. Conclusions

In this study, the influence of titanium alloy pulsed laser welding on the dynamic behavior of the molten pool is investigated, and the relationship between keyhole features and PD at different welding speeds is expounded. A method that combines KDBSCAN clustering algorithm with ES noise reduction is proposed for preprocessing keyhole features. A high-precision regression prediction model for PD is established. The main conclusions are as follows:

- (1) The PD, the width and surface area of the molten pool decreased monotonically with the increase of the welding speed, indicating that there is a correspondence between the welding speed and the geometrical features of the molten pool, which means the surface information of the molten pool is applied for predicting the PD. Based on the heat transfer mechanism, the morphology of the molten pool is directly affected by the dynamic behavior of the keyhole. Changes of keyhole features shows

consistency with that of molten pool morphology, which are able to be used to construct PD prediction models.

- (2) A pre-processing method of KDBSCAN combined with ES is proposed to reduce the apparent fluctuation of keyhole features caused by the periodic change of laser power. Compared with traditional WD and MA, the proposed method has the highest SNR, indicating that this method retains the key information in the keyhole features effectively and help to improve the quality of model input.
- (3) There is a strong positive correlation between the pre-processed feature data and the PD, and the correlation coefficient of single feature is up to 0.925. The BP model constructed based on keyhole features has the best prediction performance among multiple machine learning models, with a R^2 of 0.969, a MAE of 0.051 and RMSE of 0.088, which accurately predict the PD value.
- (4) Through the training of multiple deep learning models using raw image data, it is found that unprocessed data can be applied for PD detection likewise. Xception network exhibited the best performance, with a R^2 of 0.955, a MAE of 0.100, and a RMSE of 0.084. The results show that the choice of model and its collocation of data are crucial to the predictive effect of deep learning, which are the core factors determining the performance of the model.

The BP neural network proposed in this study shows a great performance in predicting the PD based on pre-processed keyhole features. BP neural network will be applied to the PD control during pulsed laser welding in the follow-up research, so as to provide technical support for achieving excellent welding quality. It is worth mentioning that the training results of the Xception model without manual processing are also good. The

Xception model is hopeful to show a better performance when the number of samples increases.

CRediT authorship contribution statement

Yiming Huang: Writing – review & editing, Funding acquisition. **Mingyu Li:** Writing – original draft, Formal analysis. **Kaisong Yang:** Methodology, Data curation. **Yunfu Tian:** Software, Investigation, Formal analysis. **Wei Jiang:** Visualization, Resources. **Hai Zhu:** Investigation, Conceptualization. **Haocheng Li:** Methodology, Investigation. **Zeyu Li:** Validation, Software. **Lijun Yang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under Grant No. [52201048](#) and the Natural Science Foundation of Tianjin (Grant No. [23JCQNJC01850](#)).

Appendix A. Supplementary data

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Supplementary material

References

- [1] W. Cai, J.Z. Wang, P. Jiang, L.C. Cao, G.Y. Mi, Q. Zhou

Application of sensing techniques and artificial intelligence-based methods to laser welding real-time monitoring: a critical review of recent literature

J Manuf Syst, 57 (2020), pp. 1-18, [10.1016/j.jmsy.2020.07.021](https://doi.org/10.1016/j.jmsy.2020.07.021) ↗

[View PDF](#)[View article](#)[Google Scholar](#) ↗

- [2] M. Chludzinski, R.E. dos Santos, C. Churiaque, M. Ortega-Iguña, J.M. Sánchez-Amaya
Effect of process parameters on pulsed laser welding of AA5083 alloy using response surface methodology and pulse shape variation
Int J Adv Manuf Technol, 120 (2022), pp. 4635-4646, [10.1007/s00170-022-09028-z](https://doi.org/10.1007/s00170-022-09028-z) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [3] M.J.H. Rawa, M.H. Razavi Dehkordi, M.J. Kholoud, N.H. Abu-Hamdeh, H. Azimy
Using the numerical simulation and artificial neural network (ANN) to evaluate temperature distribution in pulsed laser welding of different alloys
Eng Appl Artif Intel, 126 (2023), [10.1016/j.engappai.2023.107025](https://doi.org/10.1016/j.engappai.2023.107025) ↗
[Google Scholar](#) ↗

- [4] M. Jiang, B. Li, X. Chen, C. Tan, Z. Lei, S. Zhao, *et al.*
Numerical study of thermal fluid dynamics and solidification characteristics during continuous wave and pulsed wave laser welding
Int J Therm Sci, 181 (2022), [10.1016/j.ijthermalsci.2022.107778](https://doi.org/10.1016/j.ijthermalsci.2022.107778) ↗
[Google Scholar](#) ↗

- [5] L. Liu, H. Chen, S. Chen
Quality analysis of CMT lap welding based on welding electronic parameters and welding sound
J Manuf Process, 74 (2022), pp. 1-13, [10.1016/j.jmapro.2021.11.055](https://doi.org/10.1016/j.jmapro.2021.11.055) ↗
 [View PDF](#) [View article](#) [Google Scholar](#) ↗

- [6] L. Tomcic, A. Ederer, S. Grabmann, M. Kick, J. Kriegler, M.F. Zaeh
Interpreting acoustic emissions to determine the weld depth during laser beam welding
J Laser Appl, 34 (2022), [10.2351/7.0000796](#) ↗
[Google Scholar](#) ↗
- [7] D. Wu, Y. Huang, P. Zhang, Z. Yu, H. Chen, S. Chen
Visual-acoustic penetration recognition in variable polarity plasma arc welding process using hybrid deep learning approach
IEEE Access, 8 (2020), pp. 120417-120428, [10.1109/ACCESS.2020.3005822](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [8] B. Xue, B. Chang, D. Du
Monitoring of high-speed laser welding process based on vapor plume
Opt Laser Technol, 147 (2022), [10.1016/j.optlastec.2021.107649](#) ↗
[Google Scholar](#) ↗
- [9] Y. Huang, S. Hou, S. Xu, S. Zhao, L. Yang, Z. Zhang
EMD- PNN based welding defects detection using laser-induced plasma electrical signals
J Manuf Process, 45 (2019), pp. 642-651, [10.1016/j.jmapro.2019.08.006](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [10] M.F.M. Yusof, M. Ishak, M.F. Ghazali
Weld depth estimation during pulse mode laser welding process by the analysis of the acquired sound using feature extraction analysis and artificial neural network
J Manuf Process, 63 (2021), pp. 163-178, [10.1016/j.jmapro.2020.04.004](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [11] C. Stadter, M. Schmoeller, L. von Rhein, M.F. Zaeh

Real-time prediction of quality characteristics in laser beam welding using optical coherence tomography and machine learning

J Laser Appl, 32 (2020), [10.2351/7.00000077](#) ↗

[Google Scholar](#) ↗

- [12] S. Liu, D. Wu, Z. Luo, P. Zhang, X. Ye, Z. Yu
Measurement of pulsed laser welding penetration based on keyhole dynamics and deep learning approach

Measurement (Lond), 199 (2022), [10.1016/j.measurement.2022.111579](#) ↗

[Google Scholar](#) ↗

- [13] X. Tang, P. Zhong, L. Zhang, J. Gu, Z. Liu, Y. Gao, *et al.*
A new method to assess fiber laser welding quality of stainless steel 304 based on machine vision and hidden Markov models

IEEE Access, 8 (2020), pp. 130633-130646, [10.1109/ACCESS.2020.3009321](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [14] R. Lu, H. Wei, F. Li, Z. Zhang, Z. Liang, B. Li
In-situ monitoring of the penetration status of keyhole laser welding by using a support vector machine with interaction time conditioned keyhole behaviors

Opt Lasers Eng, 130 (2020), [10.1016/j.optlaseng.2020.106099](#) ↗

[Google Scholar](#) ↗

- [15] Z. Zhang, B. Li, W. Zhang, R. Lu, S. Wada, Y. Zhang
Real-time penetration state monitoring using convolutional neural network for laser welding of tailor rolled blanks

J Manuf Syst, 54 (2020), pp. 348-360, [10.1016/j.jmsy.2020.01.006](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [16] P. Zhao, R. Lu, J. Wu, Y. Zhang

Monitoring method of weld penetration in laser keyhole welding irradiated by laser auxiliary illuminant

J Laser Appl, 32 (2020), [10.2351/7.0000104](#) ↗

[Google Scholar](#) ↗

- [17] Y. Zhang, T. Liu, B. Li, Z. Zhang
Simultaneous monitoring of penetration status and joint tracking during laser keyhole welding

IEEE/ASME Trans Mechatronics, 24 (2020), pp. 1732-1742, [10.1109/TMECH.2019.2916984](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [18] G. He, X. Gao, L. Li, P. Gao
OCT monitoring data processing method of laser deep penetration welding based on HDBSCAN

Opt Laser Technol, 179 (2024), [10.1016/j.optlastec.2024.111303](#) ↗

[Google Scholar](#) ↗

- [19] B. Zhang, K.M. Hong, Y.C. Shin
Deep-learning-based porosity monitoring of laser welding process

Manuf Lett, 23 (2020), pp. 62-66, [10.1016/j.mfglet.2020.01.001](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [20] W. Cai, P. Jiang, L.S. Shu, S.N. Geng, Q. Zhou
Real-time monitoring of laser keyhole welding penetration state based on deep belief network

J Manuf Process, 72 (2021), pp. 203-214, [10.1016/j.jmapro.2021.10.027](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [21] R. Yu, Y. Cao, H. Chen, Q. Ye, Y.M. Zhang
Deep learning based real-time and in-situ monitoring of weld penetration: where we are and what are needed revolutionary solutions?

J Manuf Process, 93 (2023), pp. 15-46, [10.1016/j.jmapro.2023.03.011](#) ↗

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [22] H. Li, H. Ren, Z. Liu, F. Huang, G. Xia, Y. Long
In-situ monitoring system for weld geometry of laser welding based on multi-task convolutional neural network model

Measurement (Lond), 204 (2022), [10.1016/j.measurement.2022.112138 ↗](#)

[Google Scholar ↗](#)

- [23] T.Y. Liu, J.S. Bao, Zheng H. Bin, J.L. Wang, C.Q. Yang, J. Gu
Learning semantic-specific visual representation for laser welding penetration status recognition

Sci China Technol Sci, 65 (2022), pp. 347-360, [10.1007/s11431-021-1848-7 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [24] Z. Luo, D. Wu, P. Zhang, X. Ye, H. Shi, X. Cai, *et al.*
Laser welding penetration monitoring based on time-frequency characterization of acoustic emission and CNN-LSTM hybrid network

Materials, 16 (2023), [10.3390/ma16041614 ↗](#)

[Google Scholar ↗](#)

- [25] H. Lu, X. Zhang, P. Zhao, S. Bian
Effect of pulse overlap rate on the microstructure and mechanical properties of laser welds of thin AZ31 magnesium alloy

Mater Res Express, 9 (2022), [10.1088/2053-1591/ac9c49 ↗](#)

[Google Scholar ↗](#)

- [26] N. Gholizadeh, H. Saadatfar, N. Hanafi
K-DBSCAN: an improved DBSCAN algorithm for big data

J Supercomput, 77 (2021), pp. 6214-6235, [10.1007/s11227-020-03524-3 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [27] D. Arthur, S. Vassilvitskii

k-means++: the advantages of careful seeding

Proceedings of the eighteenth annual ACM-SIAM symposium on discrete algorithms (2007), pp. 1027-1035

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [28] A. Starczewski, P. Goetzen, M.J. Er
A new method for automatic determining of the DBSCAN parameters
J Artif Intell Soft Comput Res, 10 (2020), pp. 209-221, [10.2478/jaiscr-2020-0014 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [29] X. Qin, Y. Zhao, Y. Wang, J. Zeng, L. Zhou, J. Ge, *et al.*
Optimization and GRA prediction of Al-Cu pulsed laser welding process based on RSM
J Laser Appl, 36 (2024), [10.2351/7.0001249 ↗](#)
[Google Scholar ↗](#)
- [30] S. Li, G. Chen, M. Zhang, Y. Zhou, Y. Zhang
Dynamic keyhole profile during high-power deep- penetration laser welding
J Mater Process Technol, 214 (2014), pp. 565-570, [10.1016/j.jmatprotec.2013.10.019 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] L. Yang, A. Shami
On hyperparameter optimization of machine learning algorithms: theory and practice
Neurocomputing, 415 (2020), pp. 295-316, [10.1016/j.neucom.2020.07.061 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] N.B. Aldin, S.S.A.B. Aldin
Accuracy comparison of different batch size for a supervised machine learning task with image classification

2022 9th international conference on electrical and electronics engineering, ICEEE 2022,
Institute of Electrical and Electronics Engineers Inc. (2022), pp. 316-319,
[10.1109/ICEEE55327.2022.9772551](https://doi.org/10.1109/ICEEE55327.2022.9772551) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [33] N. Shirish Keskar, D. Mudigere, J. Nokedal, M. Smelyanskiy, P. Tak, P. Tang
On large-batch training for deep learning: generalization gap and sharp minima

<http://arxiv.org/abs/1609.04836> ↗ (2016)

[Google Scholar](#) ↗

- [34] F. Chollet
Xception: deep learning with depthwise separable convolutions
30th IEEE conference on computer vision and pattern recognition (2017), pp. 1800-1807

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [35] K. She, D. Li, K. Yang, M. Li, B. Wu, L. Yang, *et al.*
Online detection of laser welding penetration depth based on multi-sensor features

Materials, 17 (2024), [10.3390/ma17071580](https://doi.org/10.3390/ma17071580) ↗

[Google Scholar](#) ↗

- [36] S. Kang, M. Kang, Y. Hoon Jang, C. Kim
Spectrometer as a quantitative sensor for predicting the weld depth in laser welding

Opt Laser Technol, 175 (2024), [10.1016/j.optlastec.2024.110855](https://doi.org/10.1016/j.optlastec.2024.110855) ↗

[Google Scholar](#) ↗

[View Abstract](#)

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